

DATA 301 Assignment IV

Oliver Donaldson, 300659923

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Exercise 1

1.1 Order statistics [10%]

Wicklin’s analysis. Wicklin (2017) compares the nine sample-quantile definitions catalogued by Hyndman & Fan (1996). All nine take two adjacent order statistics $x_{(j)}, x_{(j+1)}$, with $j = \lfloor np + m \rfloor$, and return $(1 - \gamma)x_{(j)} + \gamma x_{(j+1)}$; they differ only in the offset m and in how the weight γ is set. Types 1–3 round, returning a data value or the average of two; types 4–9 interpolate linearly and can return any value in the range of the data. SAS implements five of them and defaults to type 2; R implements all nine and defaults to type 7. Plotting all nine for a ten-point sample, Wicklin finds that they differ most for small samples, near large gaps between data values and in the tails, and hardly at all for 100 or more observations away from gaps. His conclusion is that the choice matters for small samples, so the definition used should be reported; otherwise most analysts accept their software’s default. On his ten-point sample the nine sample IQRs range from 2.00 to 3.25.

Which type I would use: type 8. Of Hyndman & Fan’s (1996) six desirable properties, only type 5 has all six, but they call it a compromise that no estimation argument justifies. Types 6–9 have five each, and of these they recommend type 8 because it is approximately median-unbiased whatever the underlying distribution, so the quartiles add no reliance on the exponential model beyond the constant $\ln 3$. The type also matters for this estimator, which is the reciprocal of an IQR and so is driven by samples with a small IQR. At $n = 5$, type 7 gives $\text{IQR} = X_{(4)} - X_{(2)}$, the sum of two independent exponential spacings. Its density rises only linearly from zero, so $\mathbb{E}[\text{IQR}^{-2}] = \infty$ and type-7 $\hat{\lambda}_{\text{IQR}}$ has infinite MSE. In the simulation of Section 1.3 its estimated MSE/λ^2 comes out as 22, 42, 38 at the three values of λ instead of being equal, and a single sample supplies up to 29% of it. Type 8 interpolates between $X_{(1)}, X_{(2)}$ and $X_{(4)}, X_{(5)}$, so its IQR is a weighted sum of four spacings and its MSE is finite. At $n = 30$ and $n = 100$ type 7 still has 25% and 9% more MSE than type 8.

Default. R 4.5.2 uses type 7 (in `quantile()` and `IQR()`). All computations here set type 8 explicitly.

1.2 Bootstrap bias-corrected estimator [20%]

The bias $b(\lambda) = \mathbb{E}_\lambda[\hat{\lambda}_{\text{IQR}}] - \lambda$ depends on the unknown distribution F . The bootstrap replaces F by the empirical distribution $\hat{F}_n$ of the observed sample $\mathbf{x}$ (Robert & Casella, 2009, §1.5), which gives the bias estimate

$$\hat{b} = \mathbb{E}_*[\hat{\lambda}_{\text{IQR}}(\mathbf{X}^*)] - \hat{\lambda}_{\text{IQR}}(\mathbf{x}) \approx \frac{1}{B} \sum_{b=1}^B \hat{\lambda}_b^* - \hat{\lambda}_{\text{IQR}}(\mathbf{x}), \quad \hat{\lambda}_b^* = \hat{\lambda}_{\text{IQR}}(\mathbf{x}_b^*),$$

where $\mathbf{x}_1^*, \dots, \mathbf{x}_B^*$ are $B = 200$ resamples of $\mathbf{x}$ drawn with replacement. Subtracting it gives the first-order bias-corrected estimator

$$\tilde{\lambda} = \hat{\lambda}_{\text{IQR}} - \hat{b} = 2\hat{\lambda}_{\text{IQR}} - \frac{1}{B} \sum_{b=1}^B \hat{\lambda}_b^*.$$

It is first-order because it subtracts a single bootstrap estimate of the bias, without iterating the correction.

Implementation (`lam_iqr`, `boot_iqr` and `lam_bc` in the `.qmd`). The type-8 quartiles are computed directly from the order statistics, and all B resamples are drawn and ordered at once. A resample with zero IQR gives $\hat{\lambda}^* = \infty$; at $n = 5$ this happens only when all five values coincide (probability $5^{-4} = 1/625$). Such resamples are left out of the average.

1.3 Monte Carlo experiment [20%]

Only $\hat{\lambda}_{\text{ML}}$ has a simple closed-form MSE; $\tilde{\lambda}$ depends on the bootstrap resamples and has none, so the comparison is made by simulation. In each of the nine cells $(\lambda, n) \in \{0.1, 1, 10\} \times \{5, 30, 100\}$, R samples are drawn from $\text{Exp}(\lambda)$ and the three estimators computed on each. The MSE is estimated by the average squared error over the R samples, with Monte Carlo standard error $s/\sqrt{R}$ (Robert & Casella, 2009, §3.2).

Number of replications.

- *For each λ .* All three estimators satisfy $\hat{\lambda}(c\mathbf{x}) = \hat{\lambda}(\mathbf{x})/c$, and $X \sim \text{Exp}(\lambda)$ is Z/λ with $Z \sim \text{Exp}(1)$, so $\text{MSE}(\lambda, n) = \lambda^2 \text{MSE}(1, n)$. The *relative* Monte Carlo error therefore does not depend on λ , and the same R serves every λ at a given n .
- *For each n .* Aiming for a relative standard error of 1%, a pilot with $R = 2,000$ at $\lambda = 1$ (`working/01-pilot-mc-error.R`) gave at most 4.7% for $n = 30$ and 3.8% for $n = 100$, so $n = 30$ needs $R \approx 2000 \times 4.7^2 \approx 44,000$. I use $R = 50,000$ in every cell; the achieved relative errors at $n \geq 30$ are 0.7–1.0% (Table 1).
- *At $n = 5$.* $s/\sqrt{R}$ is only valid when the squared error has finite variance (Robert & Casella, 2009, §3.2). For $\hat{\lambda}_{\text{ML}}$ it does (the fourth moment is finite for $n > 4$), but its exact moments give the squared error a standard deviation 6.3 times its mean, so 1% would need about 393,000 replications; its MSE is also known exactly, so the simulation only has to check the code. For $\hat{\lambda}_{\text{IQR}}$ the squared error has infinite variance at $n = 5$, and $\tilde{\lambda}$ has no finite MSE at all (below), so no R can deliver a 1% error. $R = 50,000$ is therefore kept, and the $n = 5$ standard errors are reported but not relied on.

Table 1: Monte Carlo MSE, with relative Monte Carlo standard error in parentheses; $R = 50,000$, $B = 200$. At $n = 5$ the $\tilde{\lambda}$ entries do not estimate a finite quantity.

λ	n	$\hat{\lambda}_{\text{ML}}$	$\hat{\lambda}_{\text{IQR}}$	$\tilde{\lambda}$
0.1	5	0.00574 (2.0%)	0.00925 (3.1%)	12.7 (54.6%)
0.1	30	0.000392 (0.9%)	0.000942 (1.0%)	0.00115 (1.0%)
0.1	100	0.000104 (0.7%)	0.000235 (0.8%)	0.000275 (0.7%)
1.0	5	0.569 (2.2%)	0.927 (3.1%)	436,000 (95.7%)
1.0	30	0.0389 (0.8%)	0.0936 (1.0%)	0.115 (0.9%)
1.0	100	0.0105 (0.7%)	0.0234 (0.8%)	0.0276 (0.7%)
10.0	5	60.1 (2.4%)	96.5 (2.9%)	3,670,000 (95.1%)
10.0	30	3.87 (0.8%)	9.38 (1.0%)	11.5 (0.9%)
10.0	100	1.05 (0.7%)	2.39 (0.8%)	2.79 (0.7%)

Results.

- $\hat{\lambda}_{\text{ML}}$ has the smallest MSE in every cell, 2.2–2.4 times smaller than $\hat{\lambda}_{\text{IQR}}$ for $n \geq 30$, as expected when the model is correct. Its simulated MSE agrees with the exact value $\lambda^2(n+2)/[(n-1)(n-2)]$ to within 2.3 standard errors in every cell, which checks the simulation code.
- The bias correction increases the MSE. $\tilde{\lambda}$ has 23% more MSE than $\hat{\lambda}_{\text{IQR}}$ at $n = 30$ and 18% more at $n = 100$ (at every λ).
- At $n = 5$, $\tilde{\lambda}$ has no finite MSE. If $a < b$ are adjacent sample values, the resample $\{a, b, b, b, b\}$ (probability $1/625$) has type-8 IQR $(b-a)/3$ and contributes $\hat{\lambda}^* = 3 \ln 3/(b-a)$. As $b-a$ is an exponential spacing, $\mathbb{E}[1/(b-a)] = \infty$, so $\tilde{\lambda}$ has expectation $-\infty$. The Table 1 entries for $\tilde{\lambda}$ at $n = 5$ are therefore not estimates of anything: MSE/λ^2 comes out as 1,270; 436,000; 36,700 at the three values of λ , one sample supplies 52–96% of each total, and 47% of the estimates are negative. Strictly, the same argument applies at $n = 30$ and $n = 100$, but there the resamples responsible, which put both quartiles between two adjacent sample values, have probability below 10^{-12} and never occur in practice.

Conclusion. $\hat{\lambda}_{\text{ML}}$ is the best of the three throughout. The bootstrap correction does not improve $\hat{\lambda}_{\text{IQR}}$ in MSE at any sample size, and at $n = 5$ it breaks down completely.

Exercise 2

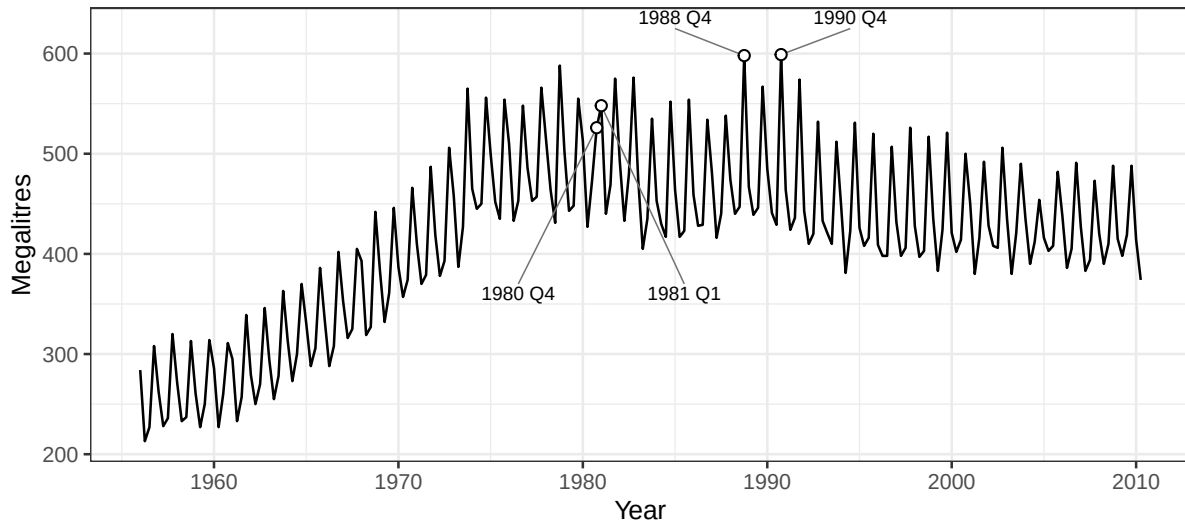


Figure 1: Quarterly beer production in megalitres, Australia, 1956 Q1 to 2010 Q2. Labelled points mark four unusual quarters.

1. Frequency. The series is quarterly: `frequency(ausbeer)` is 4, and it runs from 1956 Q1 to 2010 Q2 (218 observations). In Figure 1 the pattern repeats every four observations, with one peak and one trough a year, which is the annual cycle of quarterly data. The data show the same period in their autocorrelation function (Figure 2), the plot Hyndman & Athanasopoulos (2018, §2.8) draw for this series from 1992: the correlation peaks at lags 4, 8 and 12 (0.94, 0.89, 0.83), so each quarter is most like the same quarter a year earlier, and dips at lags 2, 6 and 10, because the Q2 trough lies two quarters from the Q4 peak.

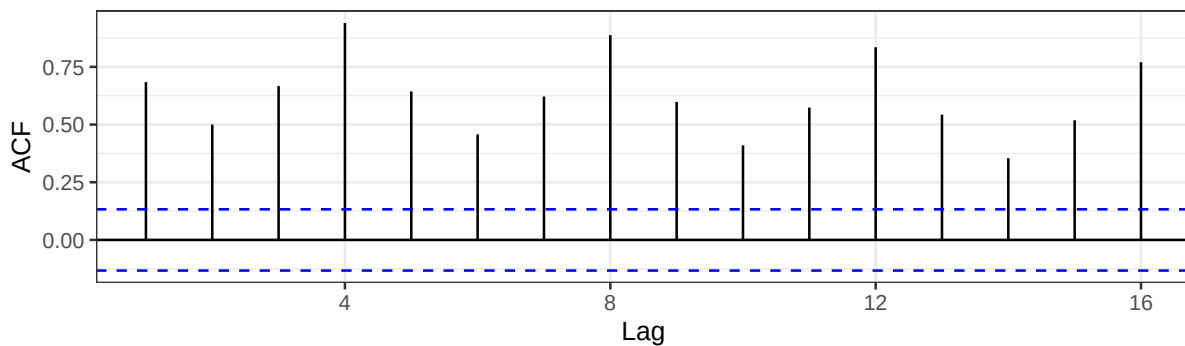


Figure 2: Autocorrelation of quarterly beer production at lags of 1 to 16 quarters, Australia, 1956 Q1 to 2010 Q2. Dashed lines mark the approximate 95% bounds for white noise.

2. Overall behaviour and trend. There is a clear long-term trend, but it is not monotone. Annual production rises from 1,032 ML in 1956 to a peak of 2,032 ML in 1981, stays high from 1974 to 1981, falls to about 1,850 ML by 1983, recovers briefly in 1988–90 and declines again to the

mid-1990s. From 1994 to 2009 it is roughly flat (1,677–1,784 ML a year) with a slight downward drift. The fall to 1983 and the 1988–90 recovery are rises and falls lasting several years with no fixed period, so they look like cyclic rather than seasonal behaviour (Hyndman & Athanasopoulos, 2018, §2.3). The trend also shows in Figure 2: every autocorrelation is positive and they decay only slowly, which is how a trend appears in an ACF (Hyndman & Athanasopoulos, 2018, §2.8).

3. Seasonality. Yes, strong seasonality. Q4 (October–December, late spring into summer, with Christmas) is the peak and Q2 (April–June, autumn into winter) the trough. The quarter means are Q4 485, Q1 417, Q3 387 and Q2 373 ML, so Q4 is about 30% above Q2, and the order $Q4 > Q1 > Q3 > Q2$ holds in 46 of the 54 complete years.

4. Is the variability constant? No. In Figure 1 the seasonal swing widens as the level rises: Q4 – Q2 averages about 88 ML in the late 1950s and 128 ML in the 1980s and 1990s. It then narrows to 94 ML in the 2000s while the level falls only 4%, so the swing is only partly proportional to the level. A transformation such as a log or Box–Cox transformation could stabilise the level-related part (Hyndman & Athanasopoulos, 2018, §3.2); Question 8 argues it is not worth doing.

5. Unusual observations. Each is compared with the same quarter in neighbouring years:

- 1981 Q1 (548 ML) is the largest Q1 in the series, 33 ML above the next (515 ML in 1978) and well above the Q1s either side (513 and 493 ML).
- 1980 Q4 (526 ML), immediately before it, is the lowest Q4 between 1973 and 1992, against 555 and 575 ML either side. A weak quarter followed by a record one suggests production shifted from one quarter into the next.
- 1988 Q4 (598 ML) and 1990 Q4 (599 ML) are the two highest values in the series, 60 and 32 ML above the previous Q4, although annual production in those years (1,959 and 1,954 ML) was below its 1981 peak.
- The step up in 1973–74 is unusual: annual production rose by 141 ML and then 79 ML, against 50–62 ML a year in 1970–72.
- The last two observations (2010 Q1–Q2) are an incomplete year: the apparent drop at the end of Figure 1 is the series stopping at the Q2 trough, not a decline.

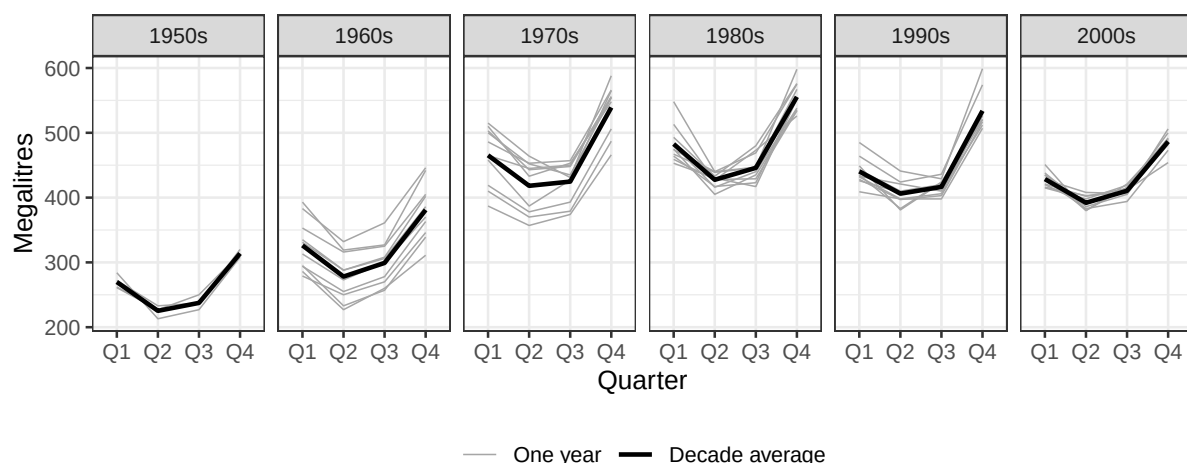


Figure 3: Quarterly beer production in megalitres by quarter and decade, Australia, complete years 1956 to 2009. Each thin line is one year; the thick line is the decade’s average for each quarter.

6. What the seasonal plot shows. Figure 3 draws each year against the quarter, one panel per decade. Every decade has the same shape: high in Q1, a dip to Q2, then a rise through Q3 to the Q4 peak. Only the height of the profile changes, rising from the 1950s to the 1980s and falling after. The seasonal pattern is therefore stable while the level changes, which is hard to see in Figure 1, where trend and season are superimposed. The panels also show the Q4 peak standing less far above the other quarters in the 2000s than in the 1970s and 1980s.

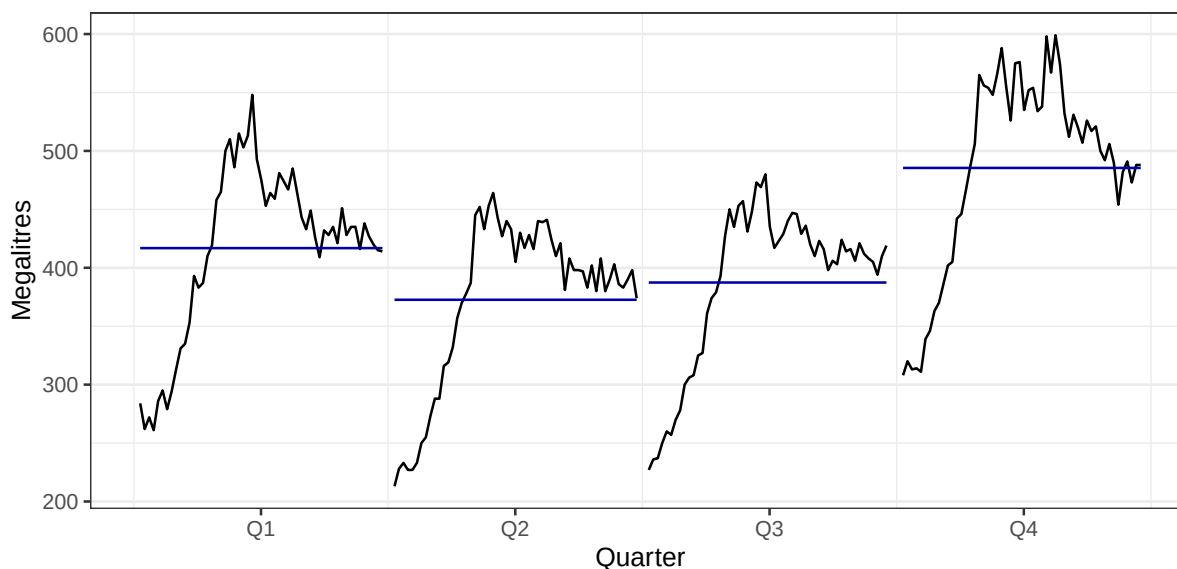


Figure 4: Quarterly beer production in megalitres by quarter, Australia, 1956 Q1 to 2010 Q2. Each panel shows one quarter in time order; the horizontal line is that quarter's mean.

7. Seasonal subseries. Q4 has the highest typical values (mean 485 ML) and Q2 the lowest (373 ML). All four quarters follow the same rise, peak and decline, so the seasonal pattern is broadly stable over time. The exception is Q4, which falls further after about 1990: 12% between 1985–94 and 2000–09, against 5–7% for the other quarters, so the seasonal amplitude shrinks over the last two decades.

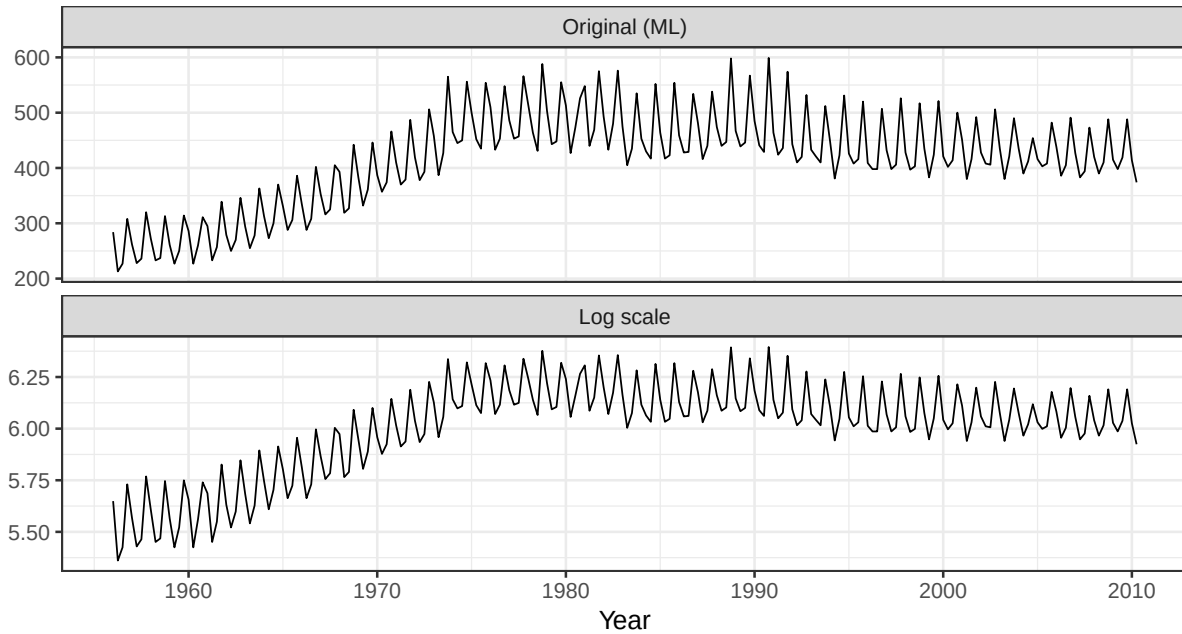


Figure 5: Quarterly beer production, Australia, 1956 Q1 to 2010 Q2, in megalitres (top) and on the log scale (bottom).

8. Detrend or transform? *Detrend: yes.* Figure 1 shows the quarterly level rising from about 260 ML in 1956 to about 500 ML around 1980 and falling back to about 430 ML, so the series has no constant mean and is not stationary. The trend should be removed, by differencing or by modelling it explicitly, before the rest of the structure is modelled.

Transform: no. A transformation is worthwhile if it makes the seasonal variation about the same size across the whole series (Hyndman & Athanasopoulos, 2018, §3.2). Figure 5 shows that the log does not do this: on the log scale the seasonal swing is widest in the 1950s–60s and narrowest at the end, so the log over-corrects the early growth and leaves the late narrowing in place. On the original scale the changes in the swing are moderate and not simply proportional to the level, so I would keep the original scale, where the values stay interpretable. As instructed, no model is fitted.

References

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- Wicklin, R. (2017). *Sample quantiles: A comparison of 9 definitions*. SAS Blogs. <https://blogs.sas.com/content/iml/2017/05/24/definitions-sample-quantiles.html>